Case 4: Zero net force, but there is motion

When a raindrop gets detached from the cloud it experiences both downward gravitational force and upward air drag force. As it descends towards the Earth, the upward air drag force increases and after a certain time, the upward air drag force cancels the downward gravity. From then on the raindrop moves at constant velocity till it touches the surface of the Earth. Hence the raindrop comes with zero net force, therefore with zero acceleration but with non-zero terminal velocity. It is shown in the Figure 3.9 (d).

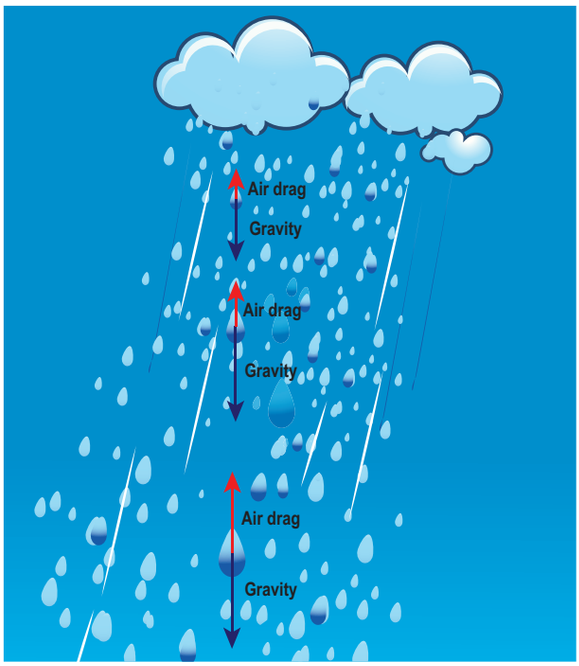
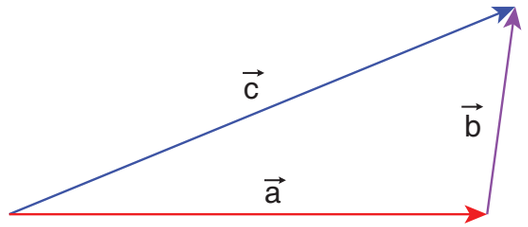


Figure 3.9 (d) Zero net force and non zero terminal velocity

4. If multiple forces $\vec{F}_1, \vec{F}_2, \vec{F}_3 \ldots . \vec{F}_n$ act on the same body, then the total force $(\vec{F}_{net})$ is equivalent to the vectorial sum of the individual forces. Their net force provides the acceleration.

$$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \ldots + \vec{F}_n$$



Vector addition of forces $\vec{a} + \vec{b}$ give resultant $\vec{c}$.

Figure 3.10 Vector addition of forces

Newton's second law for this case is

$$\vec{F}_{net} = m\vec{a}$$

In this case the direction of acceleration is in the direction of net force.

Example

Bow and arrow

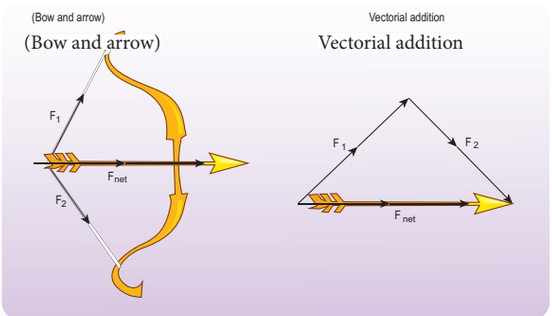


Figure 3.11 Bow and arrow – Net force is on the arrow

5. Newton's second law can also be written in the following form.
Since the acceleration is the second derivative of position vector of the body $\left(\vec{a} = \frac{d^2\vec{r}}{dt^2}\right)$, the force on the body is

$$\vec{F} = m\frac{d^2\vec{r}}{dt^2}.$$

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